

XIMB Mess Tracker

Your mess P&L, sorted. Automatically.

Imagine you eat at your college mess every day. At the end of the month, you get a big bill, but you have no idea what you actually ate or how much you spent each day. That's the problem this app solves.

XIMB Mess Tracker watches your email inbox, finds your mess invoices, reads them like a human would, and builds your daily spending diary, all by itself. You just sign in. That's it.

What Does It Do?

Think of this app like a **smart diary that writes itself**.

WHAT YOU USED TO DO MANUALLY	WHAT THE APP DOES AUTOMATICALLY
Open emails, find invoices	Scans your Gmail for invoice emails
Open each PDF, read line items	Reads the PDF and extracts every item
Write down "Dal, Rs 30, Rice, Rs 20"	Logs each item with date, name, and price
Add up daily totals on a calculator	Auto-calculates daily, monthly, and payable totals
Do this every single day	Runs every day by itself (even while you sleep)

Zero manual work. Zero missed invoices. Zero math errors.

The Automation Pipeline

This is the heart of the app. A fully automated pipeline that turns messy emails into clean financial data.

How It Works (Like Explaining to a 5-Year-Old)

- 1. You get an email.** The mess canteen sends you an invoice PDF to your Gmail.

- 2. **The app checks your mailbox.** Every day (or when you press "sync now"), it opens your Gmail and looks for emails from the mess canteen.
- 3. **It grabs the PDF attachment.** If the email has a file called `invoice-something.pdf` , it downloads it.
- 4. **It reads the PDF like a human.** The parser opens the PDF, finds the date, finds each food item and its price, and understands the table structure.
- 5. **It saves everything.** Each item (like "Paneer Butter Masala, Rs 45") gets saved to your personal database.
- 6. **You see your spending.** Open the app and your entire month's spending is laid out, day by day, item by item.

You didn't lift a finger.

Tech Stack

LAYER	TECHNOLOGY	WHY
Frontend	Next.js 16 + React 19	Fast, modern, works on phones
Styling	Tailwind CSS 4 + HeroUI	Beautiful, responsive design
Auth	Supabase Auth (Google OAuth + PKCE)	Secure sign-in with Google
Database	Supabase (PostgreSQL)	Real-time, row-level security
Gmail API	Google Gmail REST API	Reads invoice emails automatically
PDF Parser	unpdf (serverless)	Extracts text from invoice PDFs
Encryption	AES-256-GCM	Gmail tokens encrypted at rest
Hosting	Vercel (Serverless)	Auto-scales, edge-fast
Cron Jobs	Vercel Cron	Daily background sync for all users
Analytics	PostHog (self-hosted proxy)	Privacy-respecting usage tracking
Real-time	Supabase Realtime	Live updates across devices

Project Structure

```
ximb-mess-tracker/
├─ app/
│   ├─ page.tsx           # Main dashboard (the whole UI)
│   ├─ layout.tsx         # Root layout with SEO meta
│   ├─ globals.css        # Design system & styles
│   ├─ GmailSync.tsx      # Gmail auto-sync UI component
│   ├─ providers.tsx      # PostHog analytics provider
│   ├─ PostHogPageView.tsx # Page view tracking
│   └─ api/
│       └─ gmail/
│           ├─ sync/route.ts      # Sync one user's Gmail (on-demand)
│           ├─ sync-all/route.ts # Cron: sync ALL users (daily)
│           └─ store-token/route.ts # Save encrypted Gmail token
│       └─ parse-invoice/route.ts # Manual PDF upload parser
│   └─ utils/
│       ├─ gmail.ts             # Gmail API helpers (fetch, download PDFs)
│       ├─ gmailSync.ts        # Core sync engine (Gmail to DB pipeline)
│       ├─ invoiceParser.ts    # PDF text extraction & item parsing
│       ├─ crypto.ts           # AES-256-GCM encrypt/decrypt
│       ├─ supabaseAdmin.ts    # Server-side Supabase client
│       └─ supabaseClient.ts   # Client-side Supabase client
├─ vercel.json                # Cron schedule config
├─ supabase_schema.sql        # Purchases + settings tables
├─ supabase_gmail.sql         # Gmail tokens table + realtime
└─ package.json
```

Environment Variables

VARIABLE	WHAT IT DOES
NEXT_PUBLIC_SUPABASE_URL	Your Supabase project URL
NEXT_PUBLIC_SUPABASE_ANON_KEY	Public Supabase key (safe for browsers)
SUPABASE_SERVICE_ROLE_KEY	Secret Supabase key (server-only, bypasses RLS)
GOOGLE_CLIENT_ID	Google OAuth client ID
GOOGLE_CLIENT_SECRET	Google OAuth client secret
EMAIL_TOKEN_KEY	64-char hex key for encrypting Gmail tokens at rest

VARIABLE	WHAT IT DOES
CRON_SECRET	Secret for authenticating Vercel Cron jobs

Generate `GMAIL_TOKEN_KEY`

```
# Option 1: OpenSSL
openssl rand -hex 32

# Option 2: Node.js
node -e "console.log(require('crypto').randomBytes(32).toString('hex'))"
```

Getting Started

1. Clone & Install

```
git clone https://github.com/rajpal07/ximb-mess-tracker.git
cd ximb-mess-tracker
npm install
```

2. Set Up Supabase

1. Create a project at supabase.com
2. Run `supabase_schema.sql` in the SQL Editor (creates `purchases` and `settings` tables)
3. Run `supabase_gmail.sql` in the SQL Editor (creates `gmail_tokens` table + realtime)
4. Enable **Google** auth provider under Authentication > Providers
5. Copy your project URL, anon key, and service role key

3. Set Up Google OAuth

1. Go to [Google Cloud Console](https://console.cloud.google.com/)
2. Create OAuth 2.0 credentials (Web Application)
3. Add authorized redirect URI: `https://<your-supabase-project>.supabase.co/auth/v1/callback`
4. Enable the **Gmail API** in your Google Cloud project
5. Copy your client ID and client secret

4. Configure Environment

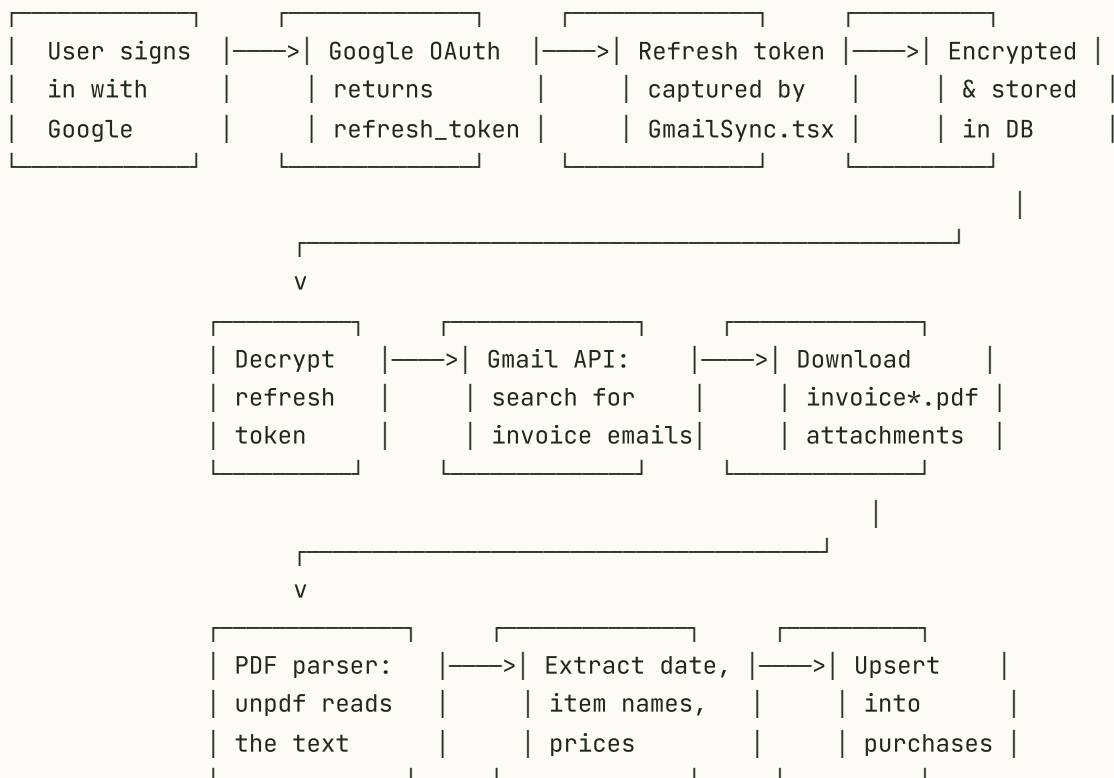
```
cp .env.local.example .env.local
# Fill in all the values from steps 2 and 3
```

5. Run Locally

```
npm run dev
```

Open <http://localhost:3000> and sign in with Google.

How Gmail Auto-Sync Works (Technical)



Three sync triggers:

- 1. On page load** - syncs when you open the app
- 2. Every 60 seconds** - polls while the tab is open
- 3. Daily cron** - Vercel Cron hits `/api/gmail/sync-all` at midnight for ALL users

Features

- **One-click Google sign-in** - No passwords, no registration forms
- **Gmail auto-sync** - Invoices are found and parsed automatically
- **Manual PDF upload** - Drag & drop invoices if you prefer
- **Manual entry** - Add custom items by hand
- **Daily breakdown** - See fixed costs + variable extras for each day
- **Monthly P&L** - Total spent, advance paid, payable balance
- **Mobile-first** - Works beautifully on phones with expandable cards
- **Push notifications** - Get notified when new invoices are synced
- **Real-time** - Changes sync across all your devices instantly
- **Secure** - Row-level security, encrypted tokens, PKCE auth flow

License

MIT

Built for XIMB students who are tired of doing mess math by hand.

Let the robots handle the boring stuff.